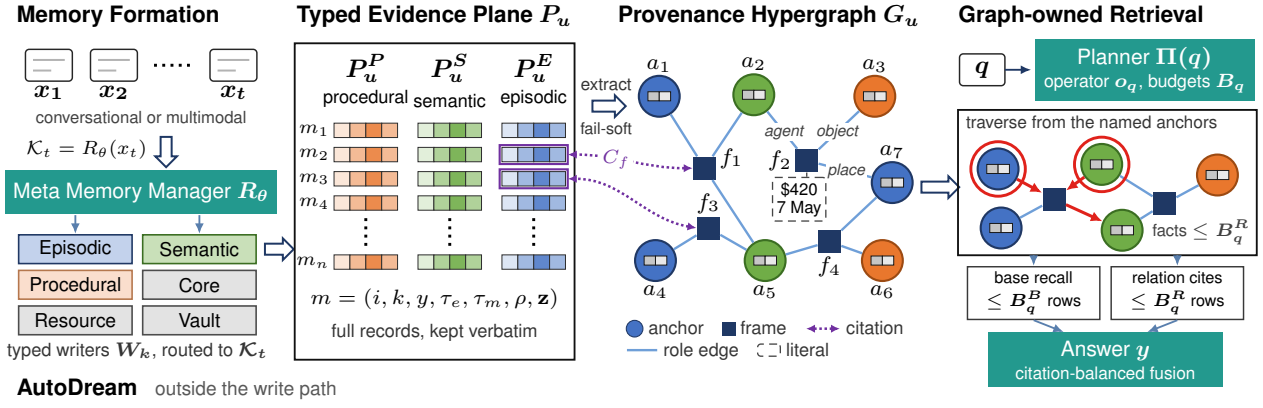
Memory Formation
x_1 x_2 ..... x_t
conversational or multimodal
$\mathcal{K}_t = R_\theta(x_t)$
Meta Memory Manager R_θ
Episodic
Semantic
Procedural
Core
Resource
Vault
typed writers W_k, routed to $\mathcal{K}_t$
Typed Evidence Plane P_u
P_u^P procedural
P_u^S semantic
P_u^E episodic
m_1
m_2
m_3
m_4
m_n
$m = (i, k, y, \tau_e, \tau_m, \rho, \mathbf{z})$
full records, kept verbatim
Provenance Hypergraph G_u
extract
fail-soft
C_f
a_1 a_2 a_3 a_4 a_5 a_6 a_7
f_1 f_2 f_3 f_4
agent
object
place
$420
7 May
anchor
frame
citation
role edge
literal
Graph-owned Retrieval
q
Planner $\Pi(q)$
operator o_q, budgets B_q
traverse from the named anchors
facts $\leq B_q^R$
base recall
$\leq B_q^B$ rows
relation cites
$\leq B_q^R$ rows
Answer y
citation-balanced fusion
AutoDream
outside the write path
Semantic memory: merge identities
$a \in \mathcal{A}_\Delta$
$b \in \mathrm{NN}_4(a)$
$\cos \geq 0.86$
same type
LLM verifier
same entity?
keeps one name
yes
no: pair cached
$a^\star$
b
$\mathbf{Dream}(P_u, G_u; \Delta_u) = (P_u, \widetilde{G}_u)$
row ids united, arms rewired
Procedural memory: learn skills
cites
Distiller
Curator
create, edit
e_1
e_2
e_m
s_1
s_2
s_k
task sessions
outcome judged from quotes
experiences
reuse
avoid
edits cite e_i
skills in P_u^P
runs once enough task sessions accumulate